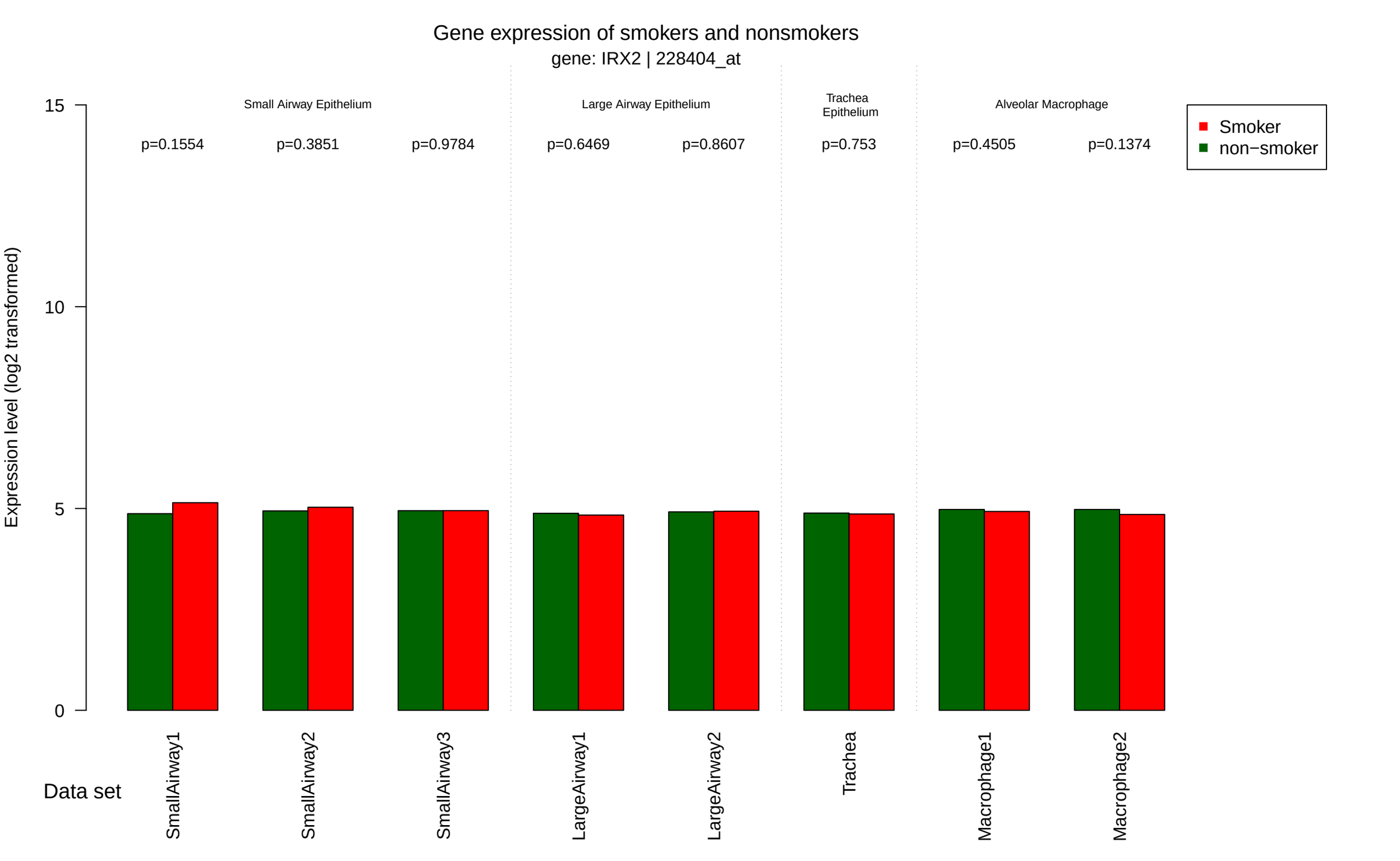
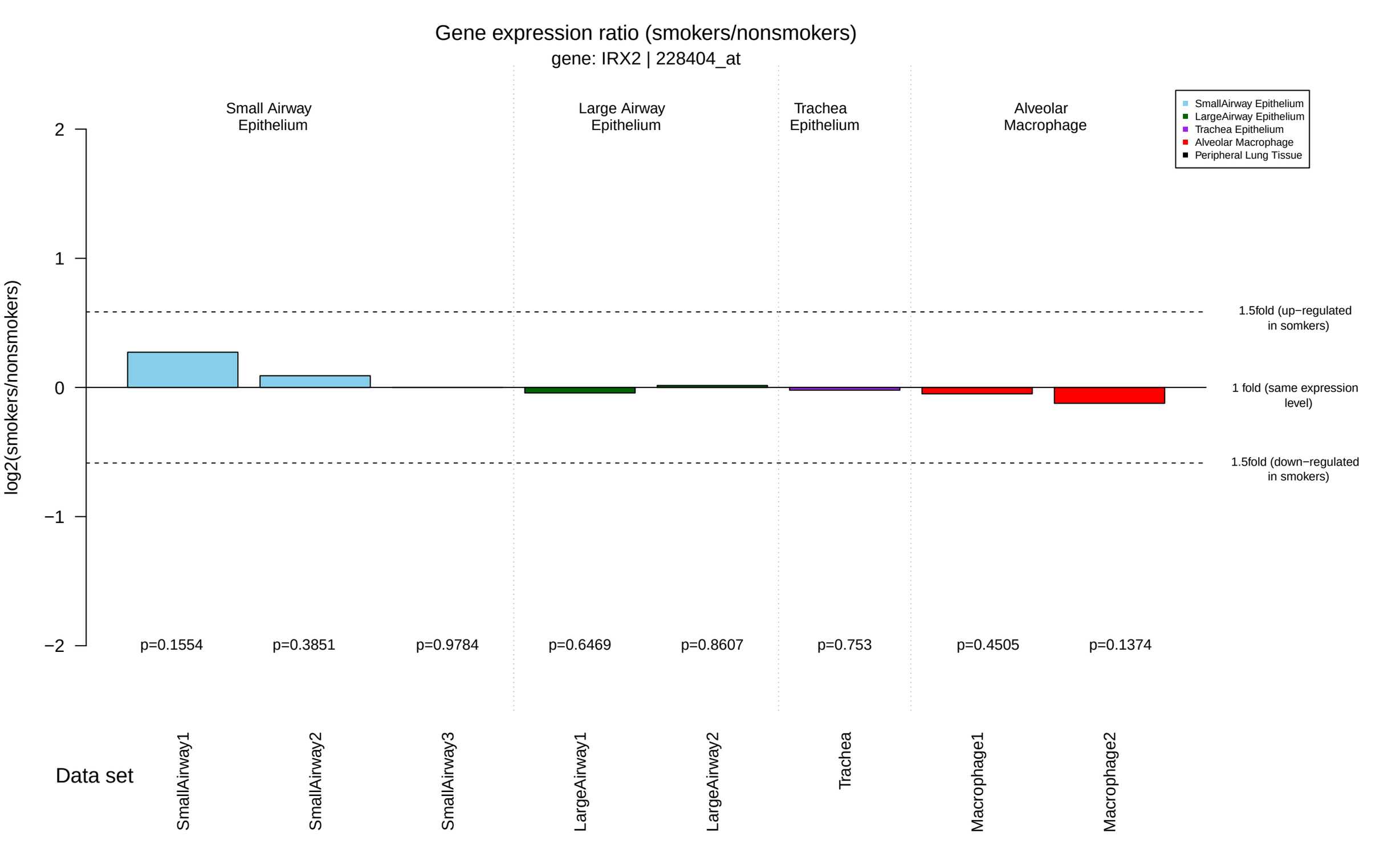
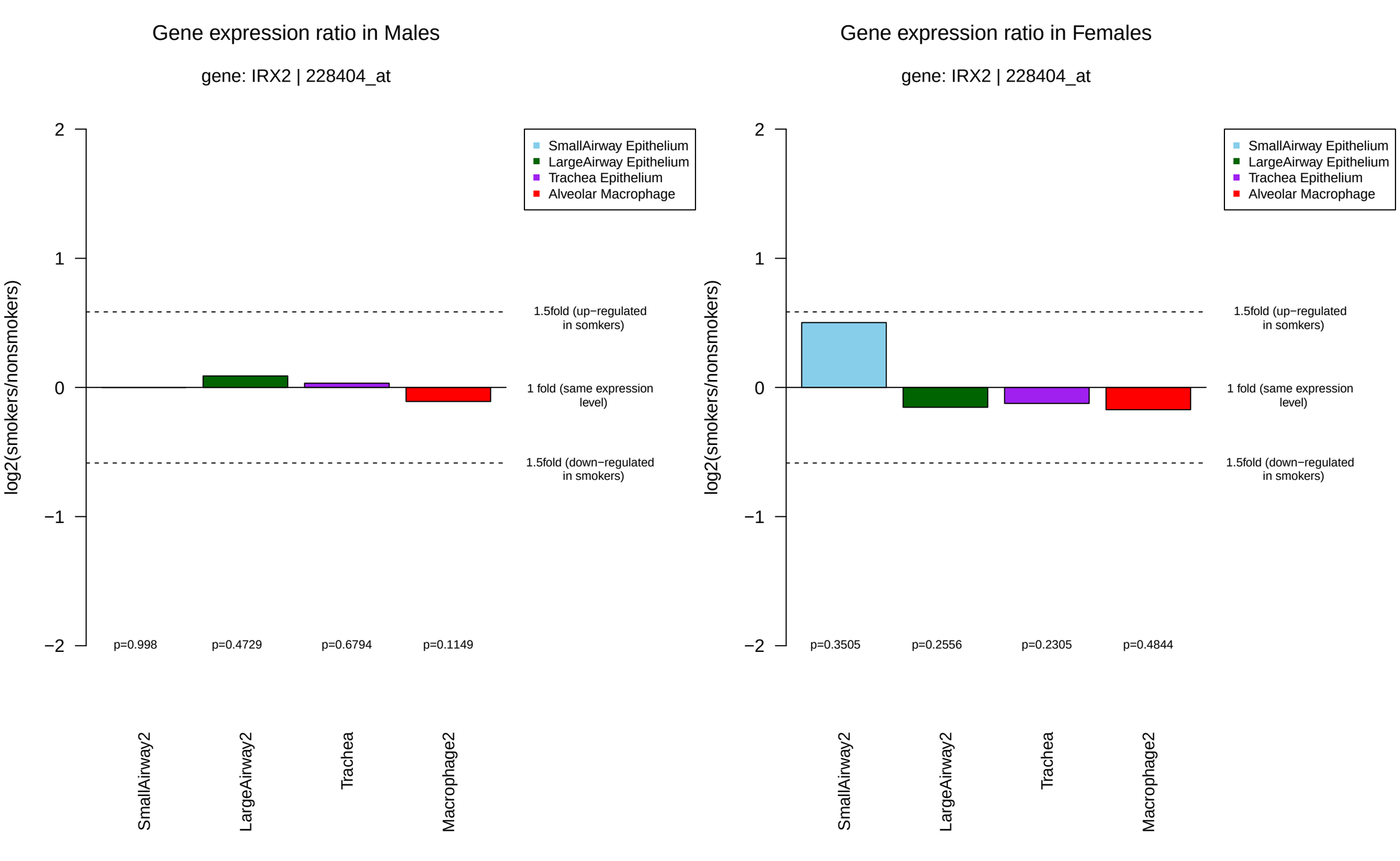
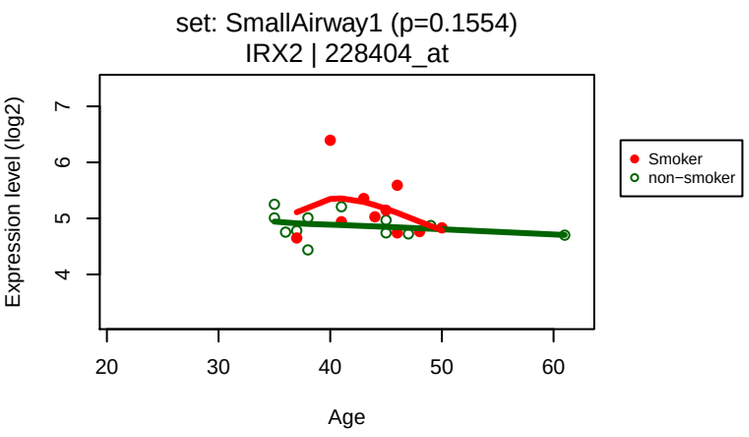
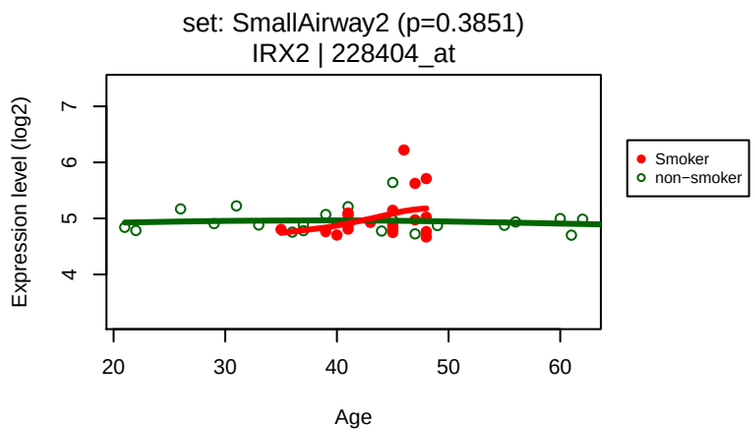
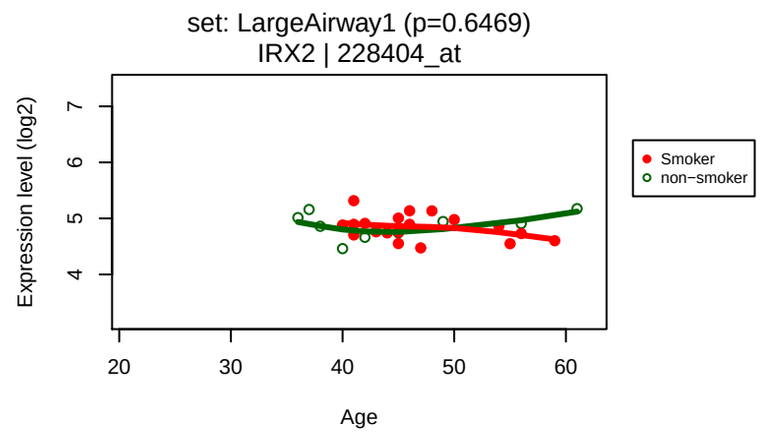

Gene expression of smokers and nonsmokers
gene: IRX2 | 228404_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1554
p=0.3851
p=0.9784
p=0.6469
p=0.8607
p=0.753
p=0.4505
p=0.1374
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

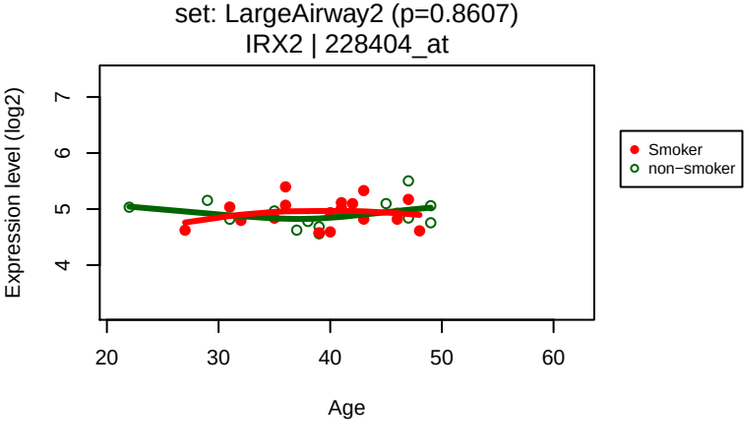

Gene expression ratio (smokers/nonsmokers)
gene: IRX2 | 228404_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1554
p=0.3851
p=0.9784
p=0.6469
p=0.8607
p=0.753
p=0.4505
p=0.1374
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

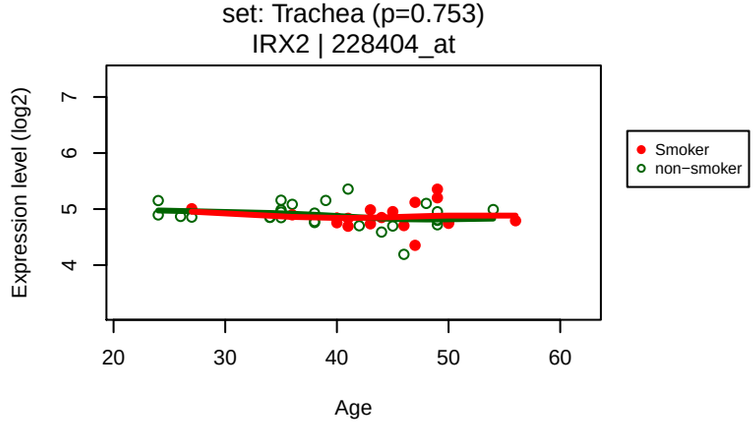

Gene expression ratio in Males
gene: IRX2 | 228404_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.998
p=0.4729
p=0.6794
p=0.1149
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRX2 | 228404_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3505
p=0.2556
p=0.2305
p=0.4844
SmallAirway2
LargeAirway2
Trachea
Macrophage2

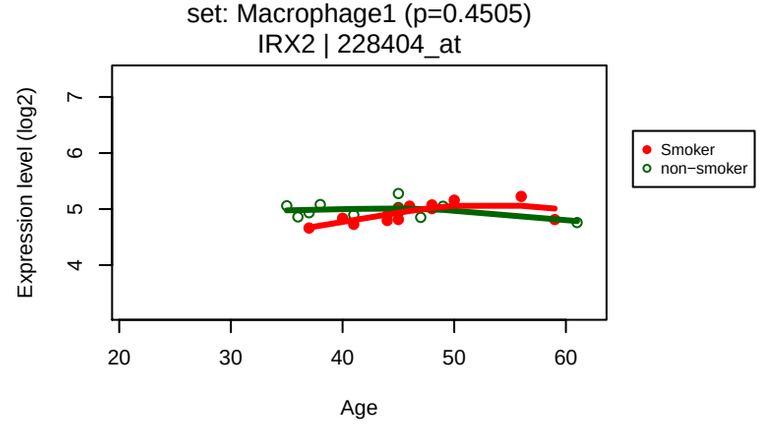

set: SmallAirway1 (p=0.1554)
IRX2 | 228404_at
Expression level (log2)
Age
Smoker
non−smoker

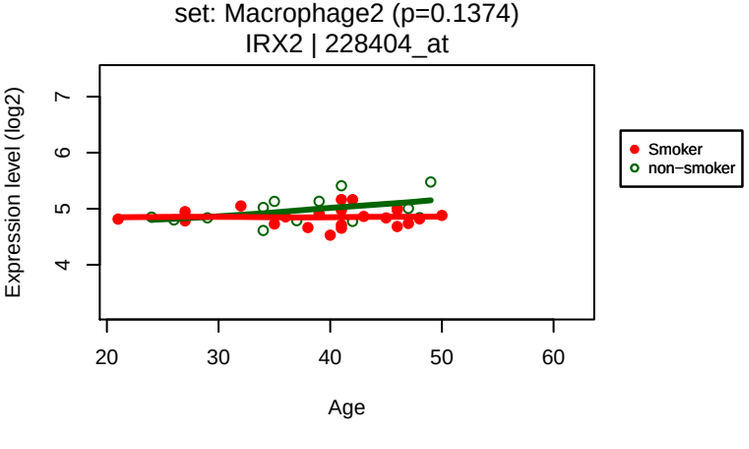

set: SmallAirway2 (p=0.3851)
IRX2 | 228404_at
Expression level (log2)
Age
Smoker
non−smoker

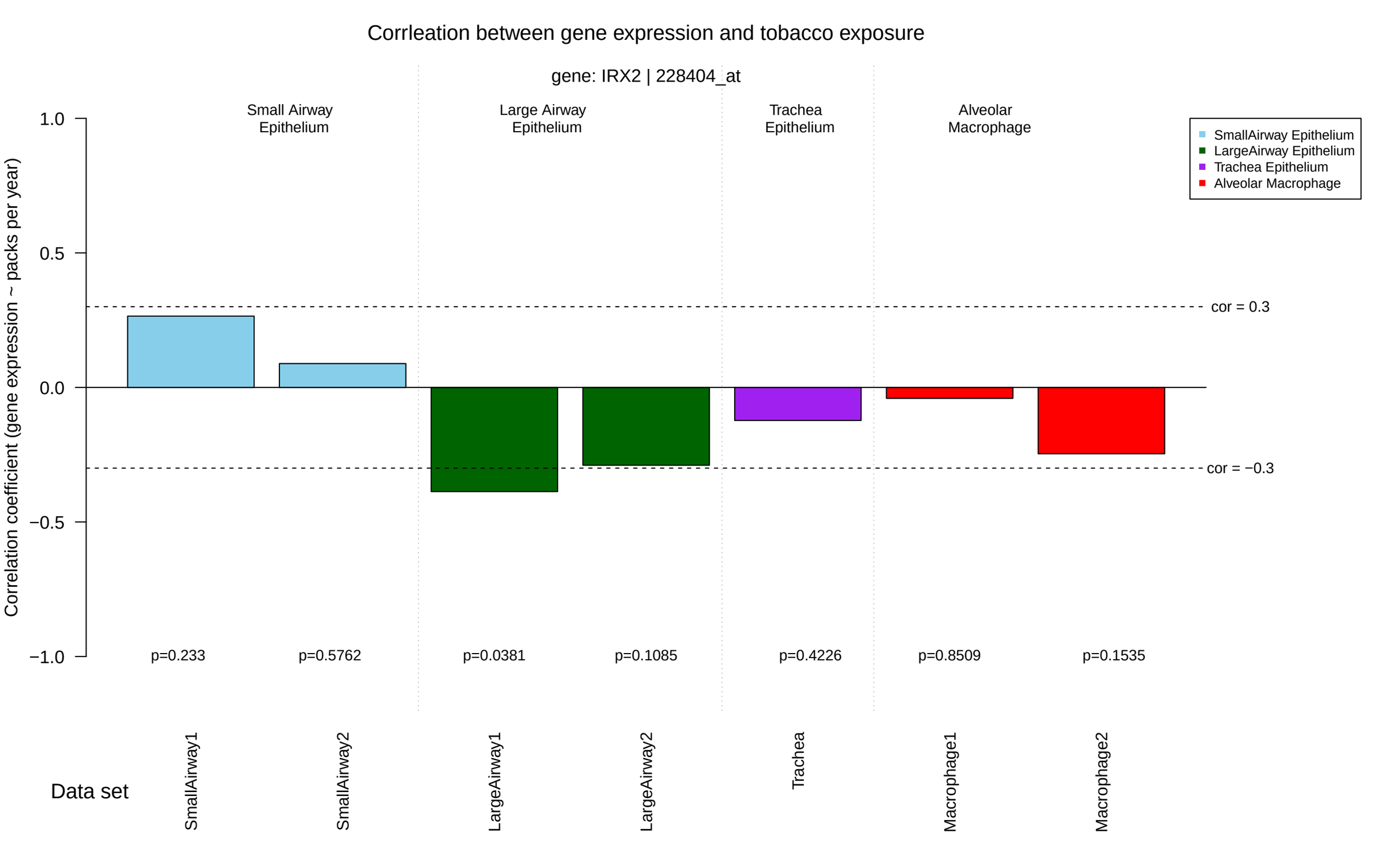

set: LargeAirway1 (p=0.6469)
IRX2 | 228404_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8607)
IRX2 | 228404_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.753)
IRX2 | 228404_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4505)
IRX2 | 228404_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1374)
IRX2 | 228404_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRX2 | 228404_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.233
p=0.5762
p=0.0381
p=0.1085
p=0.4226
p=0.8509
p=0.1535
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2